

2. Run the Installer:

- Double-click the downloaded installer file.
- Ensure you check the box that says “Add Python to PATH”.
- Click “Install Now”.

3. Verify the Installation:

- Open the Command Prompt.
- Type `python --version` and press Enter.
- You should see the installed Python version number.

macOS:**1. Download Python:**

- Go to the official Python website (<https://www.python.org/>).
- Click on the “Downloads” tab and select the latest version for macOS.

2. Run the Installer:

- Double-click the downloaded installer file.
- Follow the installation instructions.

3. Verify the Installation:

- Open the Terminal.
- Type `python3 --version` and press Enter.
- You should see the installed Python version number.

Linux:**1. Check Existing Installation:**

- Open the Terminal.
- Type `python3 --version` and press Enter.
- If Python is not installed, follow the steps below.

2. Install Python:

- Use the package manager for your distribution. For example, on Ubuntu, type:

```
sudo apt update
sudo apt install python3
```

3. Verify the Installation:

- Type `python3 --version` and press Enter.
- You should see the installed Python version number.

Basic Python Syntax

Python's syntax is straightforward and easy to understand. Here's a brief overview of some fundamental concepts and constructs in Python.

Variables and Data Types:

Python supports various data types, including integers, floats, strings, and booleans. You can assign values to variables using the `=` operator.

```
# Integer
x = 5

# Float
y = 3.14

# String
name = "John"
```



```
# Boolean
is_student = True
```

Comments:

Comments are used to explain code and are ignored by the interpreter. Single-line comments start with `#`, while multi-line comments are enclosed in triple quotes (`'''` or `"""`).

```
# This is a single-line comment
```

```
'''
```

```
This is a
multi-line comment
'''
```

Input and Output:

You can use the `print()` function to output data and the `input()` function to get user input.

```
# Output
print("Hello, World!")

# Input
user_name = input("Enter your name: ")
print("Hello, " + user_name + "!")
```

Control Structures:

Python supports control structures like conditionals and loops.

```
# Conditional
age = 18
if age >= 18:
    print("You are an adult.")
else:
    print("You are a minor.")

# Loop
for i in range(5):
    print(i)
```

Functions:

Functions are defined using the `def` keyword and allow you to encapsulate code for reuse.